

My Notes On Context Grounding

For *context grounding*, we give a context to our model and instruct and design the model to only rely on the provided context in inference time, and we also say the model to not use its parametric knowledge.

A context is basically information or evidence about our domain and case. In clinical settings, this can be a patient's historical medical records or a specific medical database about the case that we are trying to generate an answer/diagnosis.

In this way, we are forcing the model to use the given context and not to use pretraining knowledge which may be irrelevant, misleading, or even wrong about our current patient, clinical question or case.

The History of The Term: GROUNDING

The term “grounding” itself actually comes from **cognitive science** and was first used by **Stevan Harnad** in his well-known paper “Symbol Grounding Problem” (*Harnad, Stevan. "The symbol grounding problem." Physica D: Nonlinear Phenomena 42, no. 1-3 (1990): 335-346.*) Harnad proposed humans can learn language by symbols. In more details, language learning can happen by grounding words(**symbols**) to **mental pictures** or **sensory experiences**. It is impossible to learn the language by only learning the characters of the alphabet or words/text of the new language without matching those to real-world experiences.

Grounding in AI

In AI literature, we can talk about two types of grounding;

1- Sensory/Multimodal Grounding : Here we give the AI true understanding of physical reality through image or audio files or sensors, so it knows what words actually represent. This type could directly be projected to the Harnad’s “Symbol Grounding” definition.

2- Textual/Informational/Contextual Grounding : Here we give the AI trusted source context (data) to prevent hallucinations and ensure the AI’s output is factually accurate. Here there is no physical reality, we use trusted information instead. We are grounding the model’s answer to another source, not exactly a ‘symbol’. Therefore, this type of grounding differs from the original meaning.